-: FILTER DESIGN :-

FIR   IIR

Windowing

⇒ FIR means Finite Impulse response.

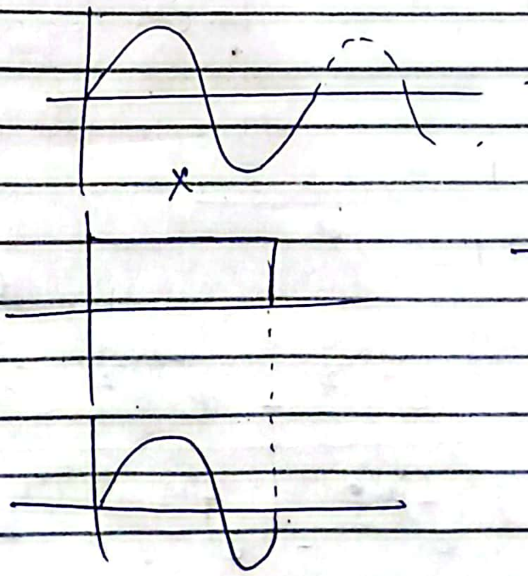
$h(n)$ has finite number of samples.

⇒ Two important characteristics :

(i) Inherently stable.
(ii) Linear phase.

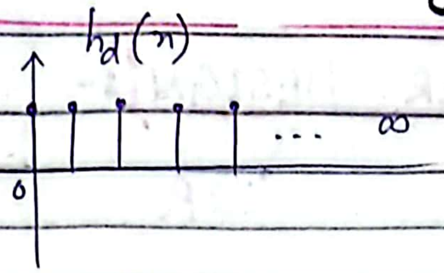
-: Windowing Method for FIR Design:-

Suppose we have a signal infinite in duration pass it through a window (means multiple it with another signal which is finite in duration.



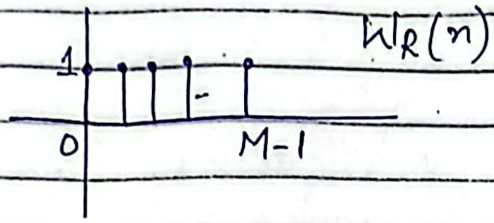
→ Shape of This Signal is rectangular so it will be rectangular window

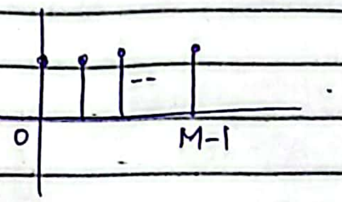
Expression for Rectangular Window :-



Desired filter response

I don't want to filter out infinite samples so filter it out with window function.





$h(n) = h_d(n) \cdot W_R(n)$

$\therefore$

$$W_R(n) = 1 \quad \text{for } 0 \leq n \leq M-1$$

$$W_R(n) = u(n) - u(n-M)$$

$$W_R(\omega) = \frac{1}{1-e^{-j\omega}} - \frac{1}{1-e^{-j\omega}} e^{-j\omega M}$$

$$= \frac{1 - e^{-j\omega M}}{1-e^{-j\omega \cancel{0}}}$$

$e^0 = 1$

$$= \frac{e^{+j\omega \frac{M}{2}} \cdot e^{-j\omega \frac{M}{2}} - e^{-j\omega \frac{M}{2}} \cdot e^{j\omega \frac{M}{2}}}{e^{j\frac{\omega}{2}} \cdot e^{-j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}} \cdot e^{-j\frac{\omega}{2}}}$$

$$W_R(\omega) = \frac{e^{-j\omega\frac{M}{2}}\left(e^{j\omega\frac{M}{2}} - e^{j\omega\frac{M}{2}}\right)}{e^{-j\frac{\omega}{2}}\left(e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}}\right)}$$

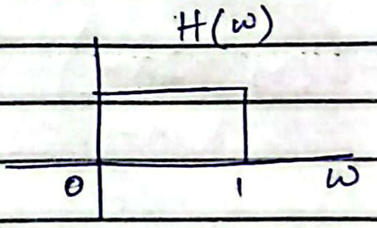
Since, $e^{j\theta} - e^{j\theta} = 2j\sin\theta$

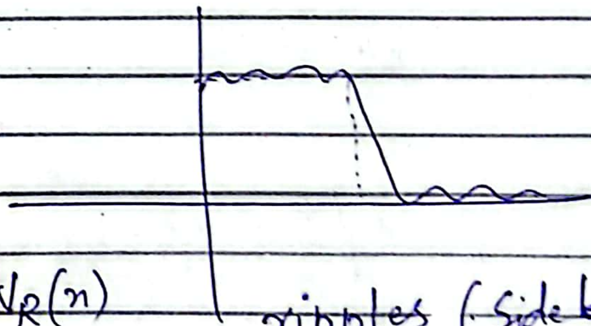
$$\therefore \quad W_R(\omega) = \frac{e^{-j\omega\frac{M}{2}}}{e^{-j\frac{\omega}{2}}} \frac{\cancel{2j}\sin\left(\frac{\omega M}{2}\right)}{\cancel{2j}\sin\left(\frac{\omega}{2}\right)}$$

$$= e^{-j\omega\frac{M}{2}} \cdot e^{j\frac{\omega}{2}} \frac{\sin\left(\frac{\omega M}{2}\right)}{\sin\left(\frac{\omega}{2}\right)}$$

$$W_R(\omega) = e^{-j\omega\left(\frac{M-1}{2}\right)} \frac{\sin\left(\frac{\omega M}{2}\right)}{\sin\left(\frac{\omega}{2}\right)}$$

Gibbs phenomena :-





$h(n) = h_d(n) \cdot W_R(n)$

ripples (Side lobe are present

$H(\omega) = H_d(n) * W_R(\omega)$

⇒ The phenomena of generation of ripples in stopband and passband is called Gibbs phenomena.

⇒ To avoid these ripples, or to reduce these oscillations, some other windows are used.

- Hamming Window
- Hanning Window
- Blackman Window.

M = order of filter.

Rectangular:

$$W_R(n) = 1$$

Hanning:-

$$W_{hn}(n) = \frac{1}{2}\left[1 - \cos\left(\frac{2\pi n}{M-1}\right)\right]$$

Hamming:-

$$W_{hm}(n) = 0.54 - 0.46\cos\left(\frac{2\pi n}{M-1}\right)$$

Triangular / Barlet:

$$W_B(n) = 1 - \frac{2|n|}{M-1}$$

$(n = 0 \text{ to } M-1 \text{ for all})$

Steps :-

(1) $H_d(\omega) = e^{-j\omega\left(\frac{M-1}{2}\right)}$

(2) Obtain $h_d(n)$ by taking IDTFT.

$$h_d(n) = \frac{1}{2\pi}\int_{-\omega c}^{\omega_c} H_d(\omega)\, e^{j\omega n}\, d(\omega)$$

(3) $h(n) = hd(n) \times W_R(n)$

$$\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

Exp 1 :- **Design linear phase filter of length 7, cutoff frequency is 1 rad/sec using rectangular window.**

Solution

$M = 7, \quad \omega_c = 1 \text{ rad/sec}$.

Step I :- $H_d(\omega) = e^{-j\omega\left(\frac{M-1}{2}\right)}$

$$= e^{-j\omega\left(\frac{7-1}{2}\right)}$$

$$H_d(\omega) = e^{-j\omega 3}$$

Step II Apply IDTFT to get $h_d(n)$

$$h_d(n) = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} H_d(\omega)\, e^{j\omega n}\, d\omega$$

$$= \frac{1}{2\pi}\int_{-1}^{1} e^{-j3\omega}\, e^{j\omega n}\, d\omega$$

$$= \frac{1}{2\pi}\int_{-1}^{1} e^{j\omega(n-3)}\, d\omega \quad -(1)$$

$$= \frac{1}{2\pi}\left.\frac{e^{j\omega(n-3)}}{j(n-3)}\right|_{-1}^{1}$$

$$= \frac{1}{2\pi}\cdot\frac{1}{j(n-3)}\left[e^{j(n-3)} - e^{-j(n-3)}\right]$$

$$= \frac{1}{\pi(n-3)}\left[\frac{e^{j(n-3)} - e^{-j(n-3)}}{2j}\right]$$

$$\therefore \theta = n-3$$

$$h_d(n) = \frac{1}{\pi(n-3)}\sin(n-3) \quad -(2)$$

* It is not valid for $n=3$

$\therefore$ for $n=3$, you have to use eq. 1

$$= \frac{1}{2\pi}\int_{-1}^{1} e^{j\omega(3-3)}\, d(\omega)$$

$$= \frac{1}{2\pi}\int_{-1}^{1} 1 \cdot d\omega$$

$$= \frac{1}{2\pi}\left[1-(-1)\right] = \frac{1}{\pi}$$

$$h_d(n) = \frac{1}{\pi} \quad \text{for } n=3 \text{ only.}$$

Step III :-

n	$h_d(n) = \frac{1}{\pi(n-3)}\sin(n-3)$	$W_R(n)=1$	$h(n)$
0	0.015	1	0.015
1	0.145	1	0.145
2	0.268	1	0.268
3	0.318	1	0.318
4	0.268	1	0.268
5	0.145	1	0.145
6	0.015	1	0.015

Exp 2:-

Design FIR filter using Hamming Window. for;

$$H_d(\omega) = e^{-j3\omega} \qquad -\frac{\pi}{4} < \omega_c < \frac{\pi}{4}$$

Solution

first find $M = ?$

$$H_d(\omega) = e^{-j\omega\left(\frac{M-1}{2}\right)}$$

$$\frac{M-1}{2} = 3$$

$$\boxed{M = 7}$$

Step I: Already given

Step II:

$$h_d(n) = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} H_d(\omega)\, e^{j\omega n}\, d(\omega)$$

$$= \frac{1}{2\pi}\int_{-\pi/4}^{\pi/4} e^{-j3\omega} \cdot e^{j\omega n}\, d\omega$$

$$= \frac{1}{2\pi}\int_{-\pi/4}^{\pi/4} e^{j\omega(n-3)}\, d\omega \quad \text{—— (1)}$$

$$= \frac{1}{2\pi} \left. \frac{e^{j\omega(n-3)}}{j(n-3)} \right|_{-\pi/4}^{\pi/4}$$

$$= \frac{1}{\pi(n-3)} \cdot \frac{1}{2j} \left(e^{j\frac{\pi}{4}(n-3)} - e^{-j\pi/4(n-3)} \right)$$

$$= \frac{1}{\pi(n-3)} \sin\left(\frac{\pi}{4}(n-3)\right)$$

not valid for $n=3$.

$\therefore$ for $n=3$.

$$hd(n) = \frac{1}{2\pi} \int_{-\pi/4}^{\pi/4} 1 \cdot d\omega$$

$$= \frac{1}{2\pi} \left[\pi/4 - (-\pi/4) \right]$$

$$= \frac{1}{\cancel{2\pi}} \frac{\cancel{2\pi}}{4}$$

$$hd(n) = \frac{1}{4} \quad \text{for } n = 3.$$

Step III

n	$hd(n) = \frac{1}{\pi(n-3)} \sin\frac{\pi}{4}(n-3)$	$W_R(n) = 0.54 - 0.46\cos\left(\frac{2\pi n}{M-1}\right)$	$h(n) = hd(n) \times W_R(n)$
0	0.075	0.08	0.006

H.W

1. Design FIR filter using rectangular window with

$$H_d(\omega) = \begin{cases} e^{-j2\omega} & -\pi/4 \le \omega \le \pi/4 \\ 0 & \pi/4 \le |\omega| \le \pi \end{cases}$$

2.

$$H_d(e^{j\omega}) = \begin{cases} 1 & 2 \le |\omega| \le \pi \\ 0 & \text{o.w.} \end{cases}$$

3.